FLYPRICE PREDICTOR

Team Members
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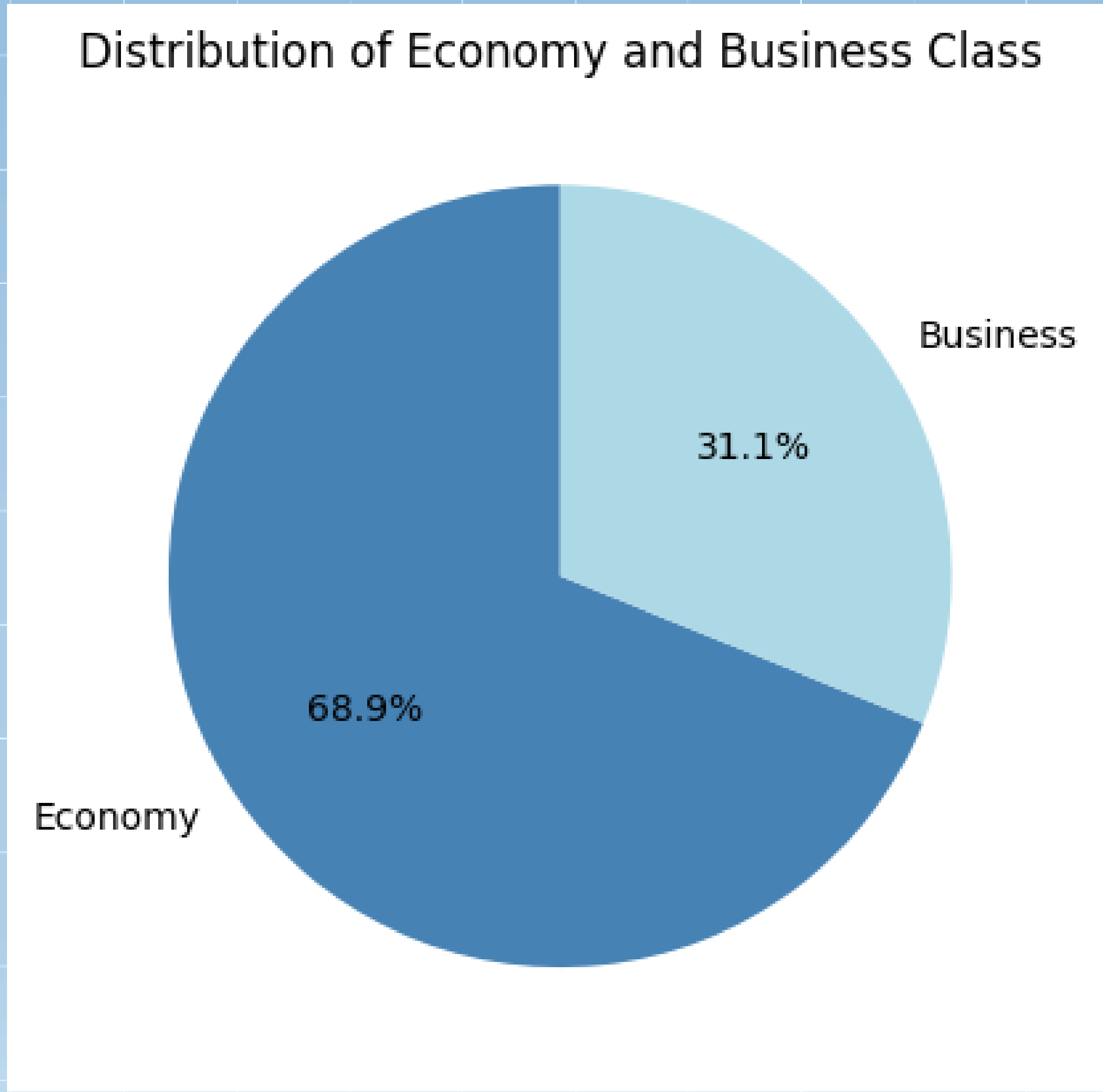
OUR DATA$ET

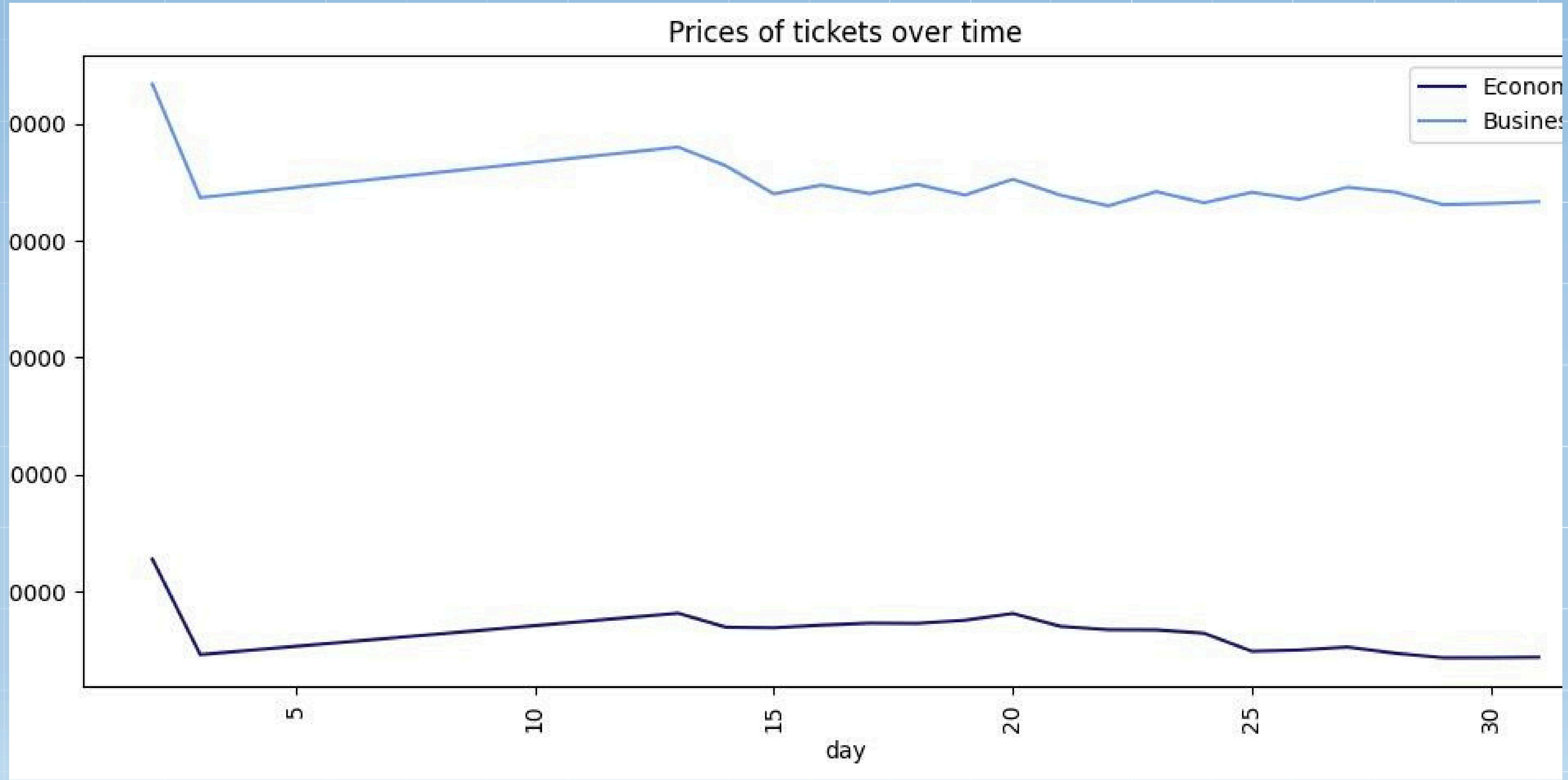
DATA SOURCE : Easemytrip

Price	Price of the ticket & target variable
Airline	Names of airlines: Air India, Air Asia, Spicejet, Vistara,Indigo, GO First
From	Departure Cities (Bangalore, Delhi, Mumbai, Kolkata, Chennai)
Stop	No. of stops between source and destination cities.
To	Destination city of the flight
Class	Information on seat class (Business & Economy)

Day	The day of the month on which the flight operated
Departure Hour	The hour of the day on which the flight departed (24hr format)
Arrival Hour	The hour of the day on which the flight arrived (24hr format)

EXPLORING OUR DATA$ET





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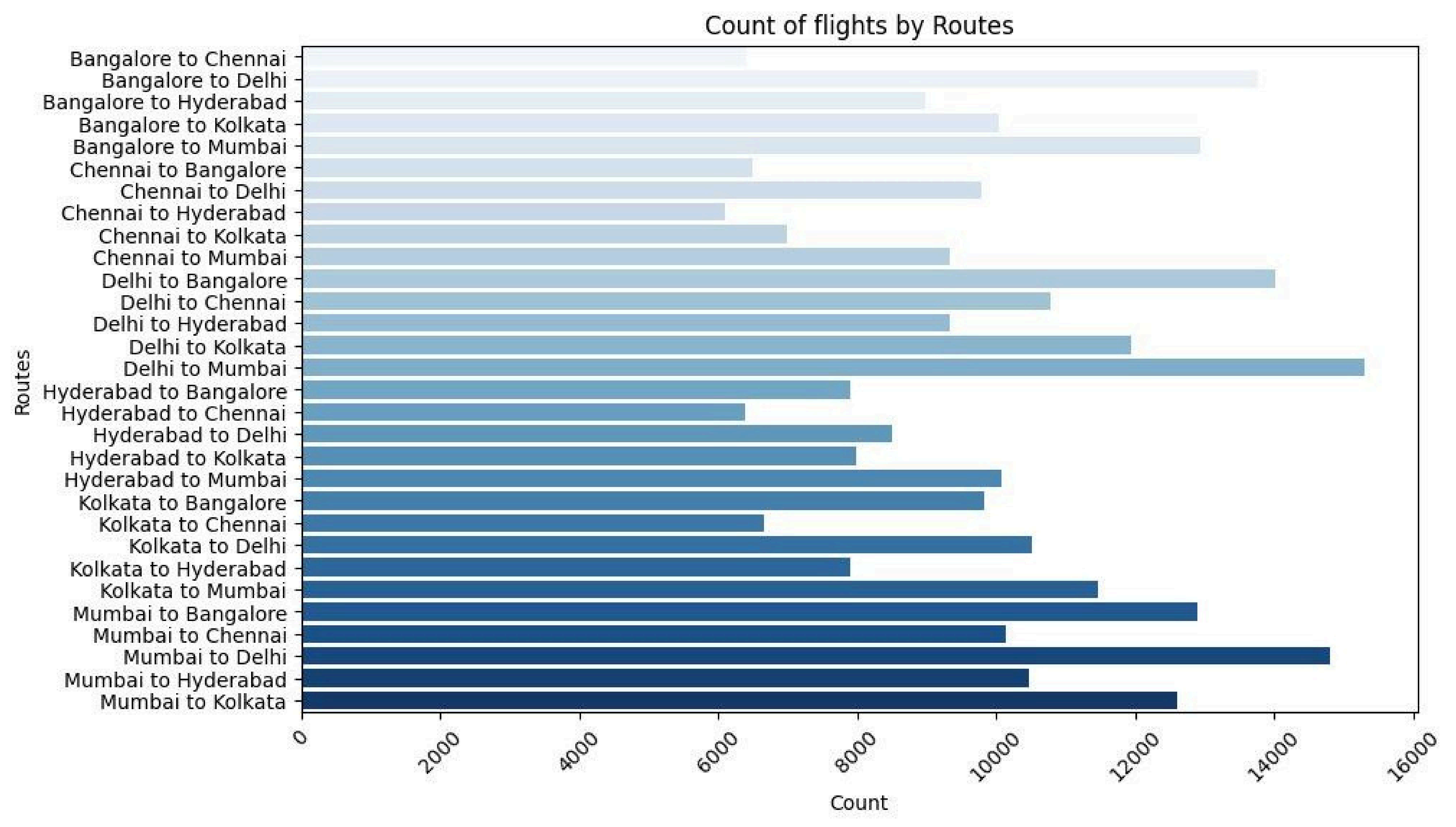
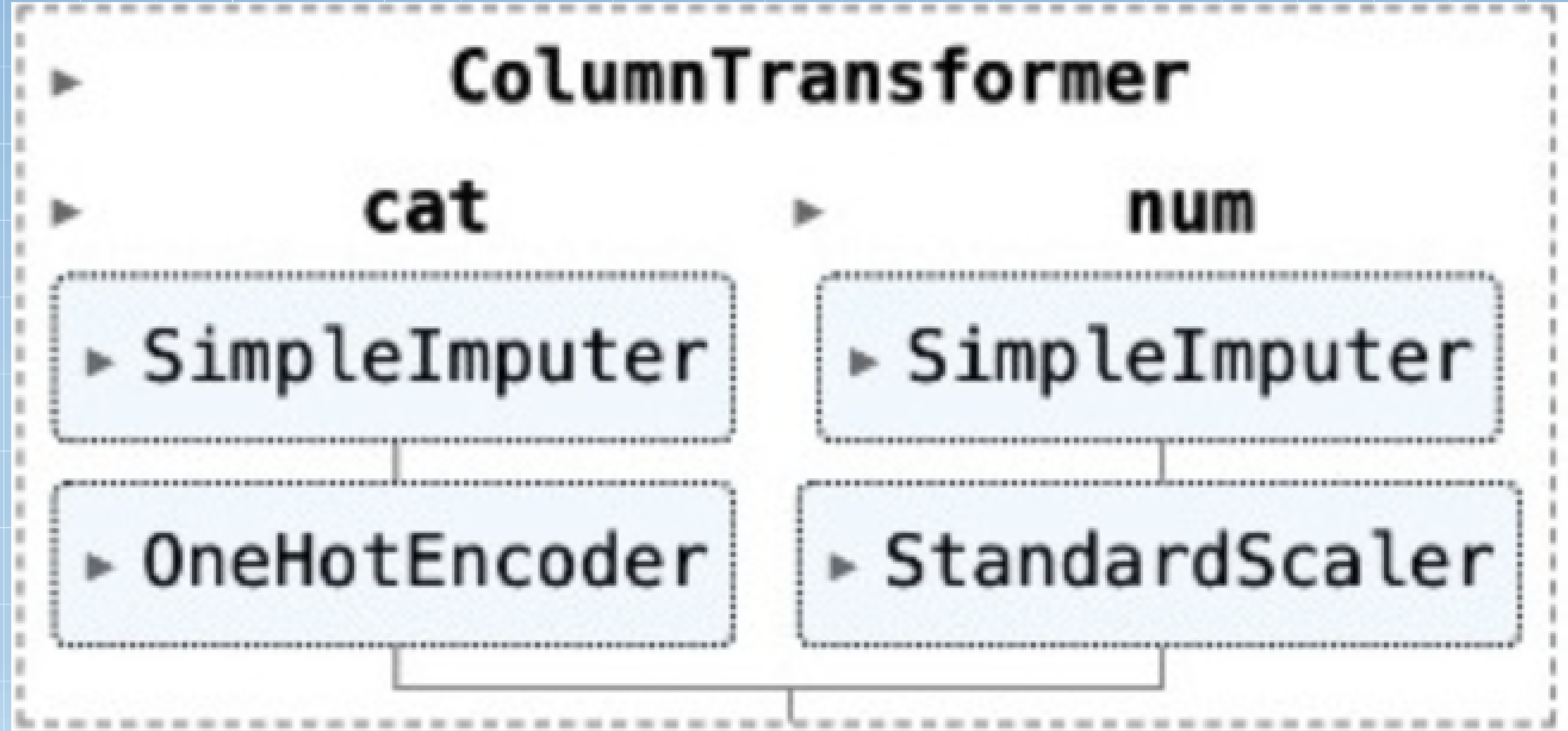


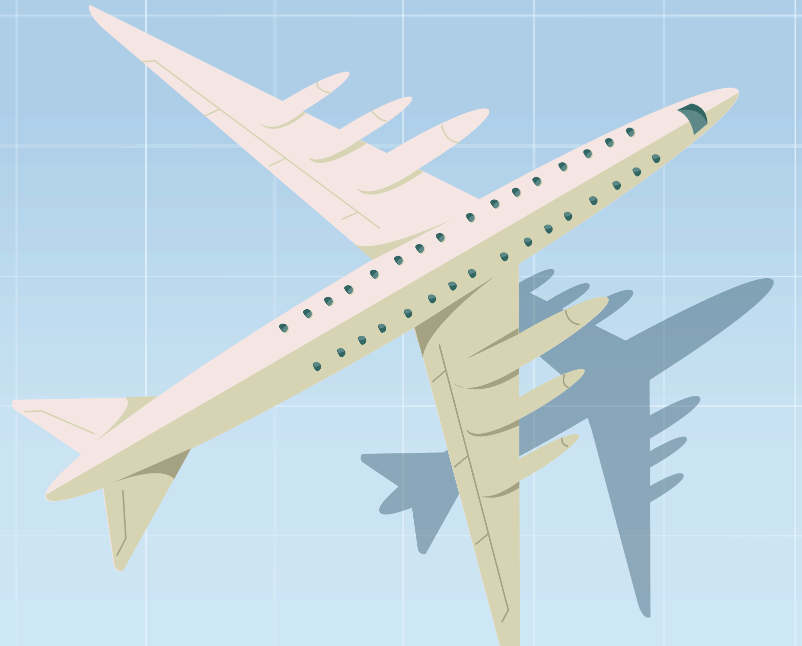
PRE-PROCE$$ING PIPELINE



TRAIN-TEST SPLIT
80-20

MODEL$ WE WORKED ON

		R-SQUARED		TUNED R-SQUARED
01	LINEAR REGRESSION	0.903		
02	RANDOM FOREST	0.969	RANDOMIZED SEARCH	0.976
03	XGBOOST	0.970	RANDOMIZED SEARCH	0.975
04	K-NN REGRESSOR	0.943		



EN$EMBLE METHOD$

R-SQUARED

01	VOTING	Multiple models make predictions on an average of individual model's output	0.968
02	STACKING	Multiple models work together—base models make predictions, and a meta-model combines these predictions for improved performance	0.973

$ELECTING OUR MODEL

- We chose to proceed with XGBoost from our two leading models, Random Forest and XGBoost Regressor.

- Hyp

	RANDOM FOREST	XGBOOST
ACCURACY	0.969 -> 0.976	0.970 -> 0.975
COMPUTATION TIME	10 MIN	3 MIN

learning_rate: 0.2

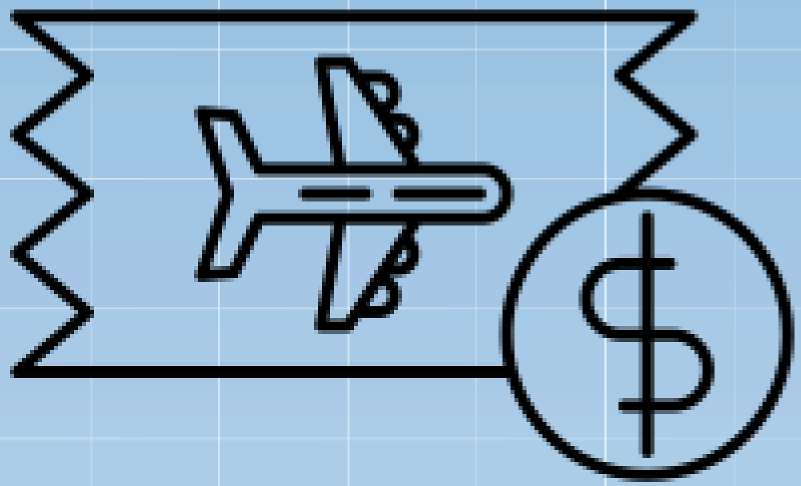
colsample_bytree: 1.0

subsample: 0.9

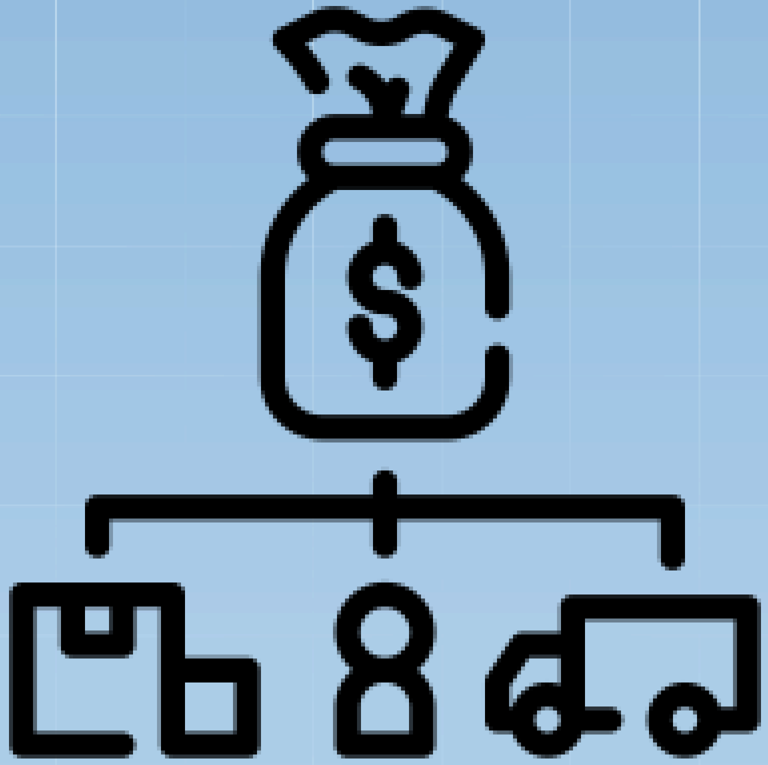
max_depth: 9

n_estimators: 64

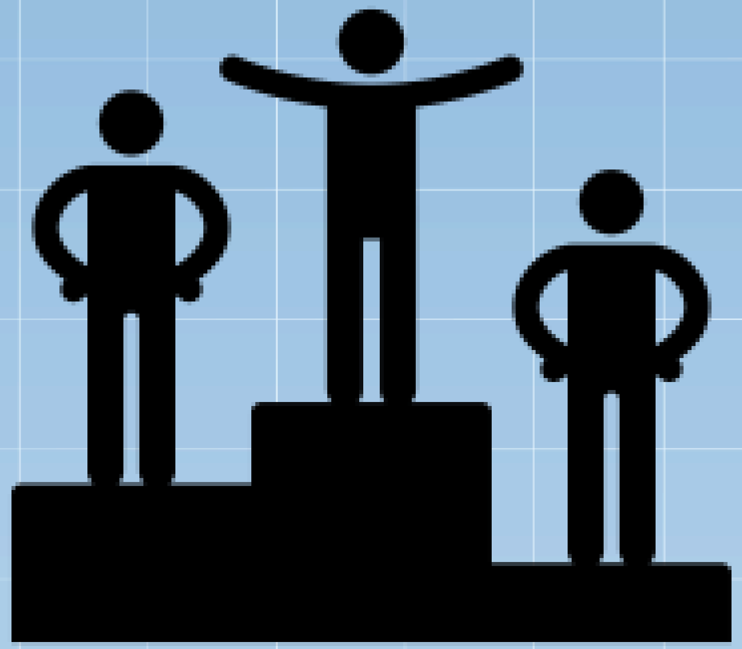
CONCLUION / BUINESS IMPLICATION



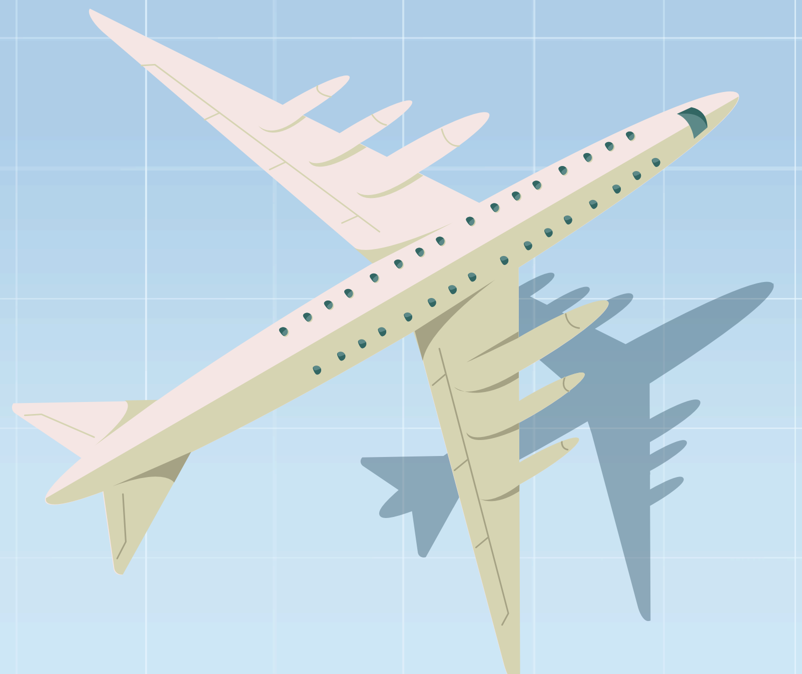
Dynamic Pricing



Optimized Cost Structure



Competitive Edge



CHALLENGE$

- Optimized Parameters for computational purpose

- Conversion of datetime variables to numerical features

- Disregarded SVM model despite a positive R-Squared after tuning

THANK$ FOR FLYING WITH U$!

NOTEBOOK:

BA810-A09-FlyPrice_Predictor